

THE 5G, IoT & SECURITY TRIANGLE

When it comes to bringing the IoT onto 5G, benefits and risks are in a tough battle to outweigh each other

Arnab Mukherjee | arnab@digit.in



The term Internet of Things was first used in 1999, at the time denoting a network that not only connects people but also the objects around them. Now, the Internet of Things, which is comprised of a vast network of interconnected electronic devices that talk to each other and take decisions by analysing that communication, is all around us. The smart objects themselves aren't new, it is the inter-object communication that is interesting and forms the premise of IoT. The installed base of the Internet of Things is forecast to reach a figure between 6.6 billion and 30 billion by 2020. If recent updates are anything to go by, it would do so with 5G connectivity in place.

WHY DOES IOT NEED 5G?

According to IHS, The number of connected devices installed worldwide has grown from 15 billion in 2015 to 26 billion in 2019, and will almost double to 51 billion in the next five years. So, yes, the IoT market is growing. A few things that 5G offers are high data rates, reduced latency, energy savings, cost reductions, and higher system capacity. It's not difficult to see how these will benefit IoT systems, but here's a quick rundown:

- **High data rates:** More data can be sent to a large number of IoT devices or singular IoT hubs.
- **Reduced latency:** IoT functionality would work seamlessly, and interdevice communication will not hamper device features.
- **Energy savings:** IoT functionality would not add on to the electricity costs.
- **Cost Reductions:** Overall cost reductions would reduce the entry barrier to IoT systems.
- **Higher system capacity:** With the number of IoT devices scaling up rapidly, the network they're on needs to be ready to handle them.
- **Reliability:** Since the 5G networks will operate on a Radio Access Architecture, with multiple endpoints closer to the end user instead of fewer cellular towers, in case a particular endpoint fails, another would be in close proximity to take over the connection.
- **Mobility:** Due to the compact nature of 5G endpoints, it would be easier to place them in locations that wouldn't be usually accessible, providing a truly mobile network that wouldn't require IoT devices on it to stay within a previously limited range.
- **Availability:** Due to the same compactness, as well as higher capac-

ity, 5G systems will have higher availability in terms of time as well as area, making them more suitable for IoT use cases.

Different combinations of these factors favour different requirements. For instance, the high data rates and low latency will favour mobile VR, whereas mobile broadband will benefit more from better availability and mobility. We need to keep in mind that a widespread 5G rollout isn't happening immediately. According to OpenSignal, the fastest average download speed in India last November on 4G was from Airtel at approximately 9Mbps. In contrast, Singapore offered an average 4G speed of 44.3Mbps back in Q4 2017. So, in India, we're still pretty much grappling with how to get 4G right. However, that doesn't exactly mean nations with better infrastructure will get 5G anytime soon either. According to a handy guide from Lifewire (<https://digit.in/5GDates>), except for a few business use cases that are either already live or being worked on, most of the world isn't getting 5G anytime before 2020.

BENEFITS OF 5G ON IOT VERTICALS

To break down the impact of 5G on IoT further, one needs to get their definition of IoT straight. While first gen IoT at home and on an industrial level has